Probability: conditioning and trees

Lecture and worked examples

Sample space, outcomes and events

A random experiment has several possible results, without it being known in advance which will occur. An outcome is a possible result; the sample space Ω is the set of them. An event is a subset of Ω; an elementary event contains a single outcome. The impossible event is $\varnothing$; the certain event is Ω.

Sample space, outcomes and events

$\overline{A}$ is the complementary event of A. $A \cap B$ means "A and B"; $A \cup B$, "A or B", possibly both. Two events are mutually exclusive if $A \cap B = \varnothing$; otherwise, they are compatible.

Sample space, outcomes and events

A six-sided die is rolled. Write Ω, A: "get an even number", B: "get 4", then $\overline{A}$, $A \cap B$ and $A \cup B$. Are A, B compatible?

Worked answer: Sample space, outcomes and events

$\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$, $B = \{4\}$, $\overline{A} = \{1, 3, 5\}$, $A \cap B = \{4\}$, $A \cup B = A$. The events are compatible because they share the outcome 4.

Sample space, outcomes and events

With $A = \{a, b, c, d, e\}$ and $B = \{b, e, f, g, h\}$, give the intersection and the union. Can $\overline{A}$ be determined without specifying the sample space?

Worked answer: Sample space, outcomes and events

$A \cap B = \{b, e\}$ and $A \cup B = \{a, b, c, d, e, f, g, h\}$. The complement depends on the sample space; it cannot be determined without this information.

Probability model and equally likely outcomes

To each outcome, a model assigns a probability that is positive or zero; the sum is 1. The probability of an event is the sum of the probabilities of its outcomes. We have

$$P(\Omega) = 1, \quad P(\varnothing) = 0, \quad P(\overline{A}) = 1 - P(A).$$

With equally likely outcomes, all outcomes have the same probability and

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}.$$

Equal likelihood is an assumption of the model, not a consequence of the list of outcomes alone.

Probability model and equally likely outcomes

Assume the die is fair. For $A =$ "even" and $C =$ "greater than or equal to 5", calculate $P(A)$, $P(C)$, $P(A \cap C)$ and $P(A \cup C)$.

Worked answer: Probability model and equally likely outcomes

$P(A) = 3/6 = 1/2$, $P(C) = 2/6 = 1/3$, $A \cap C = \{6\}$ so $P(A \cap C) = 1/6$. $A \cup C = \{2, 4, 5, 6\}$, so $P(A \cup C) = 4/6 = 2/3$. The outcome 6 must not be counted twice.

Probability model and equally likely outcomes

A drawing pin can land point up or point down. Can we deduce that each outcome has probability $1/2$? How can a model be proposed?

Worked answer: Probability model and equally likely outcomes

No: two outcomes are not enough to impose equal likelihood. One can repeat the throws under comparable conditions and use the observed frequencies to propose a model, bearing in mind fluctuations and the characteristics of the object.

Conditional probability from a table

For $P(A) > 0$, the probability of B given A is

$$P_A(B) = \frac{P(A \cap B)}{P(A)}, \qquad P(A \cap B) = P(A)P_A(B).$$

For an equally likely draw within a finite population:

$$P_A(B) = \frac{\text{Card}(A \cap B)}{\text{Card}(A)}.$$

We also write $P(B \mid A)$. $P_A(B)$ and $P_B(A)$ are generally not equal.

Conditional probability from a table

A student is drawn uniformly from the class of 32 students studied earlier. G: "boy", N: "takes CS".

Worked answer: Conditional probability from a table

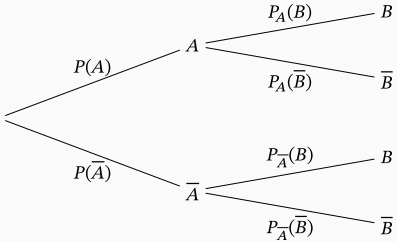
$P(G) = 18/32 = 9/16$, $P(N) = 21/32$, $P(G \cap N) = 12/32 = 3/8$. $P_N(G) = 12/21 = 4/7$, whereas $P_G(N) = 12/18 = 2/3$. By the formula: $P_N(G) = (12/32)/(21/32) = 12/21$. Given that the student takes CS, the reference population is reduced to the 21 students concerned.

Conditional probability from a table

Conditioning requires the known event to have nonzero probability; we do not divide by zero.

Weighted trees and the probability of a path

At each node, the sum of the probabilities of the outgoing branches is 1. A second-level branch carries a conditional probability. A path represents an intersection of events; its probability is the product of the probabilities written along that path.



Weighted trees and the probability of a path

We know $P(A) = 0.6$, $P_A(B) = 0.7$ and $P_{\overline{A}}(\overline{B}) = 0.9$. Complete the other branches and calculate the probabilities of the four paths.

Worked answer: Weighted trees and the probability of a path

$P(\overline{A}) = 0.4$, $P_A(\overline{B}) = 0.3$, $P_{\overline{A}}(B) = 0.1$.

$$P(A \cap B) = 0.6 \times 0.7 = 0.42, \quad P(A \cap \overline{B}) = 0.18,$$

$$P(\overline{A} \cap B) = 0.04, \qquad P(\overline{A} \cap \overline{B}) = 0.36.$$

The sum of the four probabilities is 1, which checks the consistency of the model.

Weighted trees and the probability of a path

Justify the multiplication along the path A then B from the definition of $P_A(B)$.

Worked answer: Weighted trees and the probability of a path

Since $P_A(B) = P(A \cap B)/P(A)$ and $P(A) > 0$, multiplying by $P(A)$ gives $P(A)P_A(B) = P(A \cap B)$.

From a table to a tree

For the class of 32 students, construct a tree starting with CS / no CS, then boy / girl. Then construct the tree in the reverse order. Check, in both trees, the probability of choosing a girl who takes CS.

Worked answer: From a table to a tree

First tree: initial branches $P(N) = 21/32$, $P(\overline{N}) = 11/32$. After N: $P_N(G) = 12/21 = 4/7$, $P_N(\overline{G}) = 9/21 = 3/7$. After $\overline{N}$: $6/11$ and $5/11$. Second tree: initial branches $P(G) = 18/32 = 9/16$, $P(\overline{G}) = 14/32 = 7/16$. After G: $P_G(N) = 12/18 = 2/3$, then $1/3$. After $\overline{G}$: $P_{\overline{G}}(N) = 9/14$, then $5/14$. The required path is worth $(21/32)(9/21) = 9/32$ in the first tree and $(14/32)(9/14) = 9/32$ in the second. Changing the order preserves the probabilities of the intersections but changes the probabilities of the branches.

From a table to a tree

An urn contains 3 red balls and 2 blue balls. Two balls are drawn in succession **without replacement**. Construct the tree of colours and calculate the probability of drawing two red balls.

Worked answer: From a table to a tree

First level: R with $3/5$, B with $2/5$. After R, 2 red and 2 blue remain: branches $1/2, 1/2$. After B, 3 red and 1 blue remain: branches $3/4, 1/4$. So $P(R_1 \cap R_2) = (3/5)(1/2) = 3/10$. Using $3/5$ on the second red branch would ignore the absence of replacement.

Inverting conditioning and false positives

A batch contains 10000 components, of which 100 are defective. A test is positive for 99 of the defective components and for 495 of the non-defective components. A component is chosen uniformly. D: "defective"; T: "test positive".

1. Complete the two-way table of test / true state.
2. Calculate $P_D(T)$ and $P_T(D)$.
3. Give the number of false positives and false negatives.
4. Calculate the sensitivity $P_D(T)$ and the specificity $P_{\overline{D}}(\overline{T})$.

Worked answer: Inverting conditioning and false positives

	D	$\overline{D}$	Total
T	99	495	594
$\overline{T}$	1	9405	9406
Total	100	9900	10000

$P_D(T) = 99/100 = 99\%$. On the other hand $P_T(D) = 99/594 = 1/6 \approx 16.67\%$. A positive test therefore does not imply here a 99% probability of defect. The 495 false positives are the sound components that tested positive; the false negative is the defective component that tested negative. The sensitivity is 99%; the specificity $9405/9900 = 95\%$. The low initial proportion of defective components explains why false positives outnumber true positives despite the good sensitivity.

Inverting conditioning and false positives

"Positive among the defective" and "defective among the positives" correspond to two different conditionings. Always name the reference group before the calculation.

Simulation and the law of large numbers

When an experiment is repeated a large number of times in the same model with independent repetitions, the observed frequency of an event is, barring exceptions, close to its probability. It fluctuates: it does not necessarily get closer with each new trial.

Python

```
from random import Random

def frequences_six(n, graine=1):
    # n entier >= 1 ; modele de lancers independants
    rng = Random(graine)
    succes = 0
    frequences = []
    for k in range(1, n+1):
        if rng.randint(1, 6) == 6:
            succes = succes+1
        frequences.append(succes/k)
    return frequences
```

Plot the frequencies for $n = 1000$, with several seeds. Compare with $1/6$. Why fix the seed? Can we require the final frequency to be exactly $1/6$?

Worked answer: Simulation and the law of large numbers

The plots depend on the seed and fluctuate around $1/6$. Fixing the seed makes the simulation reproducible. With 1000 throws, a final frequency of exactly $1/6$ is even impossible, because $1000/6$ is not an integer. A nearby frequency is consistent with the model, but is not a proof of it.

Modify the program to track the frequency of the event “even result”. Which reference probability should be used?

Worked answer: Simulation and the law of large numbers

Replace the test with `rng.randint(1, 6) % 2 == 0`. The reference probability is $3/6 = 1/2$. Each trial must make a single draw; do not call the generator several times to test the same result.